Package ‘dynTools’

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Title Dynamic Modeling Utilities

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Description Utility functions for data preparation when fitting dynamic models.

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https://jeksterslab.github.io/dynTools/

BugReports https://github.com/jeksterslab/dynTools/issues

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Contents

`CheckDynData` *Check Dynamic Modeling Data*

Description

The function checks whether a data frame has the variables and structure required by the dynamic modeling utility functions.

Usage

```
CheckDynData(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  require_unique = TRUE,
  require_numeric_time = TRUE,
  require_numeric_observed = TRUE,
  require_numeric_covariates = FALSE,
  min_rows = 2L
)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.
`require_unique`	Logical. If `TRUE`, require unique `id`-`time` combinations.
`require_numeric_time`	Logical. If `TRUE`, require the time variable to be numeric.
`require_numeric_observed`	Logical. If `TRUE`, require observed variables to be numeric.
`require_numeric_covariates`	Logical. If `TRUE`, require covariates to be numeric.
`min_rows`	Positive integer. Minimum number of rows required per ID.

Value

Returns `TRUE` invisibly if all checks pass.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = rep(1:3, times = 2),
  y1 = rnorm(6),
  y2 = rnorm(6)
)
CheckDynData(
```

```
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)
```

CombineByIDTime *Combine Data Sets by ID and Time*

Description

The function appends observed variables from one data set to another data set using exact `id-time` matches. Rows in `data` define the output `id-time` structure. If an `id-time` row in `data` is not present in `append_data`, missing values are inserted for the appended observed variables.

Usage

```
CombineByIDTime(data, append_data, id, time, observed, covariates = NULL)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`append_data`	Data frame. A data frame object containing the `id` and `time` columns and the observed variables to append to `data`.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), DeleteInitialNA(), DeleteObservedAllNA(), DeltaTByID(), DetrendByID(), DiagnoseScaleByID(), DiagnosticsByID(), DropByID(), ElapsedTimeByID(), ElapsedTimeByIDCT(), FilterByID(), FilterObservedRows(), FlagDiagnosticsByID(), GetDropID(), GetObservedInitial(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), PlotByID(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveCloseTimeByID(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), ScreenByID(), SubsetByID(), SummaryByID(), TrimInitialRowsByID()

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 3, 1, 2, 3),
  cov = c(10, 10, 10, 20, 20, 20)
)

append_data <- data.frame(
  id = c(1, 1, 2),
  time = c(1, 3, 2),
  y1 = c(11, 13, 22),
  y2 = c(101, 103, 202)
)

CombineByIDTime(
  data = data,
  append_data = append_data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  covariates = "cov"
)
```

DeleteInitialNA — *Delete Initial Rows With Missing Observed Values by ID*

Description

The function removes leading rows by ID when the observed variables contain missing values. This process is repeated until the first row per ID no longer has missing observed values. Covariates are retained when selecting and sorting the data, but they are not used to determine whether an initial row should be removed.

Usage

```
DeleteInitialNA(data, id, time, observed, covariates = NULL)
```

Arguments

data Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.

id Character string. A character string of the name of the ID variable in the data.

time Character string. A character string of the name of the TIME variable in the data.

observed Character vector. A vector of character strings of the names of the observed variables in the data.

covariates Character vector. A vector of character strings of the names of the covariates in the data.

Details

This is a strict helper because the first retained row must be complete on all observed variables. For more flexible trimming, prefer `TrimInitialRowsByID()` with `min_nonmissing`.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = rep(1:2, each = 5),
  time = rep(1:5, times = 2),
  y1 = c(NA, NA, 3, 4, 5, 10, 11, 12, 13, 14),
  y2 = c(NA, 2, 3, 4, 5, 10, 11, 12, 13, 14),
  y3 = c(1, NA, 3, 4, 5, 10, 11, 12, 13, 14)
)
data

DeleteInitialNA(
  data = data,
  id = "id",
```

```
  time = "time",
  observed = paste0("y", 1:3)
)
```

DeleteObservedAllNA *Delete Rows With All Observed Variables Missing*

Description

The function removes rows where all observed variables are missing.

Usage

```
DeleteObservedAllNA(data, id, time, observed, covariates = NULL)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.

Details

Covariates are retained in the returned data, but they are not used to decide whether a row should be deleted.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), CombineByIDTime(), DeleteInitialNA(), DeltaTByID(), DetrendByID(), DiagnoseScaleByID(), DiagnosticsByID(), DropByID(), ElapsedTimeByID(), ElapsedTimeByIDCT(), FilterByID(), FilterObservedRows(), FlagDiagnosticsByID(), GetDropID(), GetObservedInitial(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), PlotByID(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveCloseTimeByID(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), ScreenByID(), SubsetByID(), SummaryByID(), TrimInitialRowsByID()

Examples

```
data <- data.frame(
  id = rep(1:2, each = 5),
  time = rep(1:5, times = 2),
  y1 = c(NA, NA, 3, 4, 5, NA, 11, 12, 13, 14),
  y2 = c(NA, 2, 3, 4, 5, NA, 11, 12, 13, 14),
  y3 = c(NA, NA, 3, 4, 5, NA, 11, 12, 13, 14)
)
data

DeleteObservedAllNA(
  data = data,
  id = "id",
  time = "time",
  observed = paste0("y", 1:3)
)
```

DeltaTByID *Summarize Time Intervals by ID*

Description

The function returns a diagnostic table describing the spacing of the time variable within each ID.

Usage

```
DeltaTByID(
  data,
  id,
  time,
  observed = NULL,
  covariates = NULL,
  tolerance = sqrt(.Machine$double.eps)
)
```

Arguments

`data`	Data frame. A data frame object.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. Optional vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. Optional vector of character strings of the names of the covariates in the data.
`tolerance`	Numeric. Tolerance used to determine whether the time intervals are regular.

Value

Returns a data frame with one row per ID.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = c(1, 2, 3, 1, 3, 4),
  y = rnorm(6)
)
data

DeltaTByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y"
)
```

`DetrendByID` *Detrend Observed Variables by ID*

Description

The function removes polynomial time trends from observed variables within each ID. Missing observed values are ignored when estimating the trend and remain missing in the detrended variables.

Usage

```
DetrendByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  degree = 1L,
  replace = FALSE,
  keep_mean = TRUE,
  prefix = "detrend",
  warn_skipped = TRUE,
  drop_skipped_ids = TRUE
)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.
`degree`	Non-negative integer. Degree of the polynomial trend. Use `degree = 0` for mean centering by ID when `keep_mean = FALSE`. If `degree = 0` and `keep_mean = TRUE`, the original observed values are returned.
`replace`	Logical. If `TRUE`, replace observed variables with detrended variables. If `FALSE`, append detrended variables to the data frame.
`keep_mean`	Logical. If `TRUE`, preserve the original within-ID mean after detrending. If `FALSE`, return ordinary residuals from the within-ID trend model.
`prefix`	Character string. Prefix for detrended variables when `replace = FALSE`.

`warn_skipped` Logical. If TRUE, warn when one or more ID-variable combinations cannot be detrended because of too few usable observations, too few unique time values, or a rank-deficient trend design.

`drop_skipped_ids` Logical. If TRUE, remove all rows for IDs with at least one observed variable that cannot be detrended.

Details

If keep_mean = TRUE, the within-ID observed mean is added back after removing the fitted time trend. This removes temporal drift while preserving each ID's observed-variable level. If keep_mean = FALSE, the output is the ordinary residual from the within-ID polynomial trend model.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), CombineByIDTime(), DeleteInitialNA(), DeleteObservedAllNA(), DeltaTByID(), DiagnoseScaleByID(), DiagnosticsByID(), DropByID(), ElapsedTimeByID(), ElapsedTimeByIDCT(), FilterByID(), FilterObservedRows(), FlagDiagnosticsByID(), GetDropID(), GetObservedInitial(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), PlotByID(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveCloseTimeByID(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), ScreenByID(), SubsetByID(), SummaryByID(), TrimInitialRowsByID()

Examples

```
data <- data.frame(
  id = rep(1:2, each = 4),
  time = rep(1:4, times = 2),
  y = rep(1:4, times = 2)
)
data

DetrendByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  degree = 1,
  keep_mean = TRUE
)

DetrendByID(
  data = data,
  id = "id",
```

```
  time = "time",
  observed = "y",
  degree = 1,
  keep_mean = FALSE
)
```

`DiagnoseScaleByID` *Diagnose Scaling by ID*

Description

The function computes ID-variable-level diagnostics before within-ID scaling. It reports within-ID variability, missingness, non-finite values, and the largest standardized value that would be produced by scaling.

Usage

```
DiagnoseScaleByID(
  data,
  id,
  time,
  observed,
  sd_min = 0.1,
  z_cut = 6,
  min_n = 3L,
  flagged_only = FALSE
)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`sd_min`	Numeric scalar or `NULL`. Minimum acceptable within-ID standard deviation before within-ID scaling. If `NULL`, low-SD flagging is skipped.
`z_cut`	Numeric scalar. Absolute standardized-value threshold used to flag potentially extreme values after within-ID scaling.
`min_n`	Positive integer. Minimum number of finite observations required within each ID-variable combination.
`flagged_only`	Logical. If `TRUE`, return only flagged ID-variable combinations. If `FALSE`, return all ID-variable combinations.

Value

Returns a data frame with one row per ID-variable combination.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), CombineByIDTime(), DeleteInitialNA(), DeleteObservedAllNA(), DeltaTByID(), DetrendByID(), DiagnosticsByID(), DropByID(), ElapsedTimeByID(), ElapsedTimeByIDCT(), FilterByID(), FilterObservedRows(), FlagDiagnosticsByID(), GetDropID(), GetObservedInitial(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), PlotByID(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveCloseTimeByID(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), ScreenByID(), SubsetByID(), SummaryByID(), TrimInitialRowsByID()

Examples

```
data <- data.frame(
  id = rep(1:2, each = 5),
  time = rep(1:5, times = 2),
  y1 = c(1, 2, 3, 4, 20, 2, 2, 2, 2, 2),
  y2 = c(5, 4, 3, 2, 1, 1, 1, 1, 1, 2)
)

DiagnoseScaleByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  z_cut = 3
)
```

DiagnosticsByID *ID-Level Diagnostics for Dynamic Modeling Data*

Description

The function computes ID-level diagnostics for intensive longitudinal data. Diagnostics include the number of finite observed rows, number of complete rows, proportion of rows with no finite observed values, duplicate ID-time rows, time gaps, non-finite observed values, within-ID standard deviations, and counts of extreme observed values.

Usage

```
DiagnosticsByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  missing_codes = NULL,
  min_nonmissing = 1L,
  extreme_cut = c(4, 5, 6),
  sd_cut = c(0.1, 0.05),
  time_scale = 1,
  posix_unit = "hours"
)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.
`missing_codes`	`NULL` or vector. Values to temporarily treat as missing when computing diagnostics. The input data are not modified.
`min_nonmissing`	Positive integer. Minimum number of finite observed variables required for a row to count as an observed row.
`extreme_cut`	Numeric vector. Absolute-value cutoffs used to count extreme values.
`sd_cut`	Numeric vector. Within-ID standard-deviation cutoffs used to count low-variance variables.
`time_scale`	Positive number. Multiplier applied to numeric time gaps. For example, use `time_scale = 24` when time is measured in days and gaps should be reported in hours.
`posix_unit`	Character string. Unit for gaps when `time` is a `POSIXt` or `Date` variable. Passed to `as.numeric.difftime()`.

Value

Returns a data frame with one row per ID.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 3, 1, 2, 3),
  y1 = c(1, NA, 3, 1, 1, 1),
  y2 = c(NA, NA, 4, 2, 2, 2)
)

DiagnosticsByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)
```

DropByID — *Drop IDs From a Data Frame*

Description

The function removes rows belonging to one or more IDs. It is a small convenience wrapper used after ID-level diagnostics identify IDs for exclusion or sensitivity analysis.

Usage

```
DropByID(data, id, drop = NULL)
```

Arguments

`data`	Data frame.
`id`	Character string. Name of the ID variable.
`drop`	Vector. ID values to remove. If `NULL` or empty, `data` is returned unchanged.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = c(1, 1, 2, 2, 3, 3),
  y = 1:6
)

DropByID(
  data = data,
  id = "id",
  drop = c(2, 3)
)
```

ElapsedTimeByID — *Create Elapsed Time by ID*

Description

The function creates an elapsed-time variable within ID. For date-time variables, elapsed time is computed with `difftime()`. For numeric time variables, elapsed time is computed by subtraction, optionally converting from `input_units` to `units`.

Usage

```
ElapsedTimeByID(
  data,
  id,
  time,
  output = "time_elapsed",
  units = "days",
  origin = c("by_id", "global"),
```

```
  input_units = NULL,
  replace = FALSE
)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
output	Character string. Name of the elapsed-time variable.
units	Character string. Units for the output elapsed time.
origin	Character string. If `origin = "by_id"`, each ID's origin is its own minimum non-missing time. If `origin = "global"`, all IDs use the global minimum non-missing time.
input_units	Character string or `NULL`. Units of numeric input time. If `NULL`, numeric time is assumed to already be in the desired output scale and only subtraction is performed.
replace	Logical. If `TRUE`, replace the original `time` variable. If `FALSE`, append `output`.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = c(1, 1, 2, 2),
  datetime = as.POSIXct(
    c(
      "2020-01-01 00:00:00",
      "2020-01-02 00:00:00",
```

```
      "2020-01-03 12:00:00",
      "2020-01-04 00:00:00"
    ),
    tz = "UTC"
  )
)
data

ElapsedTimeByID(
  data = data,
  id = "id",
  time = "datetime",
  output = "time",
  units = "days"
)
```

ElapsedTimeByIDCT *Create CT-Scaled Elapsed Time by ID*

Description

The function creates an elapsed-time variable within ID and rescales it for continuous-time state space modeling. The rescaling divides elapsed time by a typical positive consecutive time interval so that the typical `delta_t` is approximately 1.

Usage

```
ElapsedTimeByIDCT(
  data,
  id,
  time,
  output = "time_ct",
  units = "hours",
  origin = c("by_id", "global"),
  input_units = NULL,
  scale = c("mean_dt", "median_dt"),
  scale_value = NULL,
  replace = FALSE
)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.

`time`	Character string. A character string of the name of the TIME variable in the data.
`output`	Character string. Name of the elapsed-time variable.
`units`	Character string. Units for the output elapsed time.
`origin`	Character string. If `origin = "by_id"`, each ID's origin is its own minimum non-missing time. If `origin = "global"`, all IDs use the global minimum non-missing time.
`input_units`	Character string or `NULL`. Units of numeric input time. If `NULL`, numeric time is assumed to already be in the desired output scale and only subtraction is performed.
`scale`	Character string. Method used to compute the time-scaling divisor. If `scale = "mean_dt"`, the divisor is the mean positive consecutive elapsed-time interval. If `scale = "median_dt"`, the divisor is the median positive consecutive elapsed-time interval.
`scale_value`	Numeric or `NULL`. Optional user-supplied positive scaling divisor in the elapsed-time `units`. If supplied, this value is used instead of estimating the divisor from the data.
`replace`	Logical. If `TRUE`, replace the original `time` variable. If `FALSE`, append `output`.

Details

This is useful for continuous-time models because the units of time are arbitrary, but poorly scaled time can lead to drift parameters that are very large or very small.

The minimum positive consecutive interval is reported as a diagnostic but is not used as a scaling option. The goal is to make the typical consecutive interval approximately 1, not the smallest interval.

Value

Returns a data frame. The returned data frame has an attribute named `"time_ct_scale"` containing the scaling information.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = c(1, 1, 2, 2),
  datetime = as.POSIXct(
    c(
      "2020-01-01 00:00:00",
      "2020-01-01 06:00:00",
      "2020-01-03 12:00:00",
      "2020-01-04 00:00:00"
    ),
    tz = "UTC"
  )
)

data_ct <- ElapsedTimeByIDCT(
  data = data,
  id = "id",
  time = "datetime",
  output = "time_ct",
  units = "hours",
  scale = "mean_dt"
)

attr(data_ct, "time_ct_scale")
```

FilterByID *Filter Dynamic Modeling Data by ID*

Description

The function removes IDs that do not satisfy simple minimum data requirements. For diagnostic screening and sensitivity-analysis decisions, prefer `DiagnosticsByID()`, `FlagDiagnosticsByID()`, `GetDropID()`, and `ScreenByID()`.

Usage

```
FilterByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  min_rows = 2L,
  min_complete = 2L,
  max_prop_missing = 1,
  allow_initial_na = TRUE,
  allow_all_missing = FALSE
)
```

Arguments

- `data` — Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
- `id` — Character string. A character string of the name of the ID variable in the data.
- `time` — Character string. A character string of the name of the TIME variable in the data.
- `observed` — Character vector. A vector of character strings of the names of the observed variables in the data.
- `covariates` — Character vector. A vector of character strings of the names of the covariates in the data.
- `min_rows` — Non-negative integer. Minimum number of rows required per ID.
- `min_complete` — Non-negative integer. Minimum number of complete observed rows required per ID.
- `max_prop_missing` — Numeric. Maximum allowed proportion of missing values across observed variables.
- `allow_initial_na` — Logical. If `FALSE`, remove IDs where the initial row contains missing observed values. Covariates are not used to determine whether the initial row has missing values.
- `allow_all_missing` — Logical. If `FALSE`, remove IDs where all observed values are missing.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = rep(1:3, each = 3),
  time = rep(1:3, times = 3),
```

```
  y = c(1, 2, 3, NA, NA, 4, NA, NA, NA)
)
data

FilterByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  min_complete = 2
)
```

FilterObservedRows *Remove Rows With Too Few Observed Values*

Description

The function removes rows with fewer than a requested number of non-missing observed variables. This is useful after regularizing or padding intensive longitudinal data, where many rows may contain no observed measurements but retain an ID and time value.

Usage

```
FilterObservedRows(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  min_nonmissing = 1L,
  min_rows_by_id = NULL
)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.

`min_nonmissing` Positive integer. Minimum number of non-missing observed variables required to retain a row.

`min_rows_by_id` `NULL` or positive integer. If not `NULL`, IDs with fewer than `min_rows_by_id` retained rows are removed after row filtering.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2),
  time = c(1, 2, 3, 1, 2),
  y1 = c(1, NA, 3, NA, 2),
  y2 = c(NA, NA, 4, NA, 3)
)

FilterObservedRows(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)
```

`FlagDiagnosticsByID` *Flag Potentially Problematic IDs*

Description

The function adds rule-based flags to the output of `DiagnosticsByID()`. It is intended for sensitivity checks before fitting dynamic models. It flags low observation counts, missingness, duplicate ID-time rows, non-finite observed values, large time gaps, low variability, and extreme observed values.

Usage

```
FlagDiagnosticsByID(
  x,
  min_observed_rows = 30L,
  min_complete_rows = NULL,
  max_prop_all_missing = 0.95,
  max_gap = NULL,
  max_median_gap = NULL,
  min_sd = 0.05,
  extreme_cut = 6,
  drop_score_cut = 2L
)
```

Arguments

`x`	Data frame returned by `DiagnosticsByID()`.
`min_observed_rows`	Positive integer. Minimum number of observed rows expected per ID.
`min_complete_rows`	`NULL` or positive integer. Minimum number of complete rows expected per ID.
`max_prop_all_missing`	`NULL` or number between 0 and 1. Maximum tolerated proportion of all-missing observed rows.
`max_gap`	`NULL` or positive number. Maximum tolerated observed-row time gap in the units used by `DiagnosticsByID()`.
`max_median_gap`	`NULL` or positive number. Maximum tolerated median observed-row time gap in the units used by `DiagnosticsByID()`.
`min_sd`	`NULL` or non-negative number. Minimum tolerated within-ID standard deviation across observed variables. If `NULL`, low-SD flagging is skipped.
`extreme_cut`	Positive number. Absolute-value cutoff used to flag extreme observations.
`drop_score_cut`	Positive integer. IDs with priority scores greater than or equal to this value are marked as sensitivity-drop candidates.

Value

Returns `x` with flag columns appended.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, PlotByID(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveCloseTimeByID(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), ScreenByID(), SubsetByID(), SummaryByID(), TrimInitialRowsByID()

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 3, 1, 2, 3),
  y1 = c(1, NA, 3, 1, 1, 1),
  y2 = c(NA, NA, 4, 2, 2, 2)
)

diagnostics <- DiagnosticsByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)

FlagDiagnosticsByID(
  x = diagnostics,
  min_observed_rows = 3,
  min_sd = 0.05
)
```

GetDropID *Get Sensitivity-Drop IDs From Diagnostics*

Description

The function extracts IDs marked as sensitivity-drop candidates by FlagDiagnosticsByID().

Usage

```
GetDropID(x, flagged_only = TRUE)
```

Arguments

x	Data frame returned by FlagDiagnosticsByID().
flagged_only	Logical. If TRUE, return IDs marked as sensitivity-drop candidates. If FALSE, return all flagged IDs.

Value

Returns a vector of IDs.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 3, 1, 2, 3),
  y1 = c(1, NA, 3, 1, 1, 1),
  y2 = c(NA, NA, 4, 2, 2, 2)
)

diagnostics <- DiagnosticsByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)

flagged <- FlagDiagnosticsByID(
  x = diagnostics,
  min_observed_rows = 3,
  min_sd = 0.05
)

GetDropID(flagged)
```

`GetObservedInitial` *Get Observed Initial Means and Covariances*

Description

The function extracts the first observed time point for each ID and computes the sample mean vector and sample covariance matrix of the observed variables across IDs. The resulting mean vector and covariance matrix can be used as fixed initial conditions in dynamic models.

Usage

```
GetObservedInitial(data, id, time, observed, ridge = 1e-06)
```

Arguments

- `data` Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
- `id` Character string. A character string of the name of the ID variable in the data.
- `time` Character string. A character string of the name of the TIME variable in the data.
- `observed` Character vector. A vector of character strings of the names of the observed variables in the data.
- `ridge` Positive numeric. Small value added to the diagonal of the covariance matrix if the estimated covariance matrix is not positive definite.

Details

Rows are sorted by ID and time before selecting the first row for each ID. Missing observed values are ignored when computing the mean vector. The covariance matrix is computed using complete cases of the selected first-time observations.

If the covariance matrix is not positive definite, a small ridge value is added to the diagonal.

Value

Returns a list with the following elements:

- `data` Data frame containing the first observed time point for each ID and the selected observed variables.
- `mean` Numeric vector of observed-variable means computed from the first observed time point across IDs.
- `cov` Observed-variable covariance matrix computed from complete cases of the first observed time point across IDs.
- `n` Number of IDs with a first observed row.
- `n_complete` Number of IDs with complete data on all selected observed variables at the first observed time point.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = rep(1:3, each = 4),
  time = rep(1:4, times = 3),
  y1 = c(1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6),
  y2 = c(4, 3, 2, 1, 5, 4, 3, 2, 6, 5, 4, 3)
)
data

init <- GetObservedInitial(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)

init$data
init$mean
init$cov
init$n
init$n_complete
```

InitialNA — *Check for Missing Observed Values in Initial Row by ID*

Description

The function checks whether the initial row for each ID has missing values in the observed variables. Covariates are retained when selecting and sorting the data, but they are not used to determine whether the initial row is missing.

Usage

```
InitialNA(data, id, time, observed, covariates = NULL)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.

Value

Returns a vector of ID values where the initial row has any missing observed value.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = rep(1:2, each = 5),
  time = rep(1:5, times = 2),
  y1 = c(NA, NA, 3, 4, 5, 10, 11, 12, 13, 14),
  y2 = c(NA, 2, 3, 4, 5, 10, 11, 12, 13, 14),
  y3 = c(1, NA, 3, 4, 5, 10, 11, 12, 13, 14)
)
data

InitialNA(
  data = data,
  id = "id",
  time = "time",
  observed = paste0("y", 1:3)
)
```

`InsertNA` *Insert NAs for Missing Observations*

Description

The function creates a sequence of time values. This is a simple helper retained for compatibility. For new workflows, prefer `RegularizeTimeByID()`, which provides explicit global/by-ID grids and preserve/snap behavior.

Usage

```
InsertNA(data, id, time, observed, covariates = NULL, delta_t)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.
`delta_t`	Positive number. Time interval.

Details

It starts with the smallest time value as the starting point and the largest time value as the endpoint. The sequence is incremented by `delta_t`. This new sequence is combined with the existing empirical time values. For any specific time value where there are no observations, NAs are inserted.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 4, 1, 3, 4),
  y1 = c(10, 11, 13, 20, 22, 23),
  y2 = c(5, 6, 8, 15, 17, 18)
)
data

InsertNA(
```

```
    data = data,
    id = "id",
    time = "time",
    observed = c("y1", "y2"),
    delta_t = 1
  )
```

LagByID *Create Lagged Variables by ID*

Description

The function creates lagged observed variables within ID. Lagged values never cross ID boundaries.

Usage

```
LagByID(data, id, time, observed, covariates = NULL, lags = 1L, prefix = "lag")
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.
`lags`	Positive integer vector. Lags to create.
`prefix`	Character string. Prefix for the lagged variable names.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = rep(1:3, times = 2),
  y1 = rnorm(6),
  y2 = rnorm(6)
)
data

LagByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  lags = 1
)
```

`MakeClockTime` *Make Clock Time*

Description

The function combines a date and a clock time into a `POSIXct` vector. Clock time can be supplied as decimal hours, for example `13.5`, or as a character time, for example `"13:30"` or `"13:30:00"`.

Usage

```
MakeClockTime(
  date,
  time,
  tz = "UTC",
  date_formats = c("%m/%d/%y", "%m/%d/%Y", "%Y-%m-%d"),
  invalid = c("NA", "error")
)
```

Arguments

`date`	Date vector. Dates can be supplied as `Date`, `POSIXt`, or character values.
`time`	Numeric or character vector. Clock time in decimal hours, `"HH:MM"`, or `"HH:MM:SS"` format.
`tz`	Character string. Time zone passed to `as.POSIXct()`.
`date_formats`	Character vector. Date formats used to parse character dates.
`invalid`	Character string. If `invalid = "NA"`, invalid clock times are returned as `NA`. If `invalid = "error"`, invalid clock times throw an error.

Value

Returns a `POSIXct` vector.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
MakeClockTime(
  date = "2020-01-02",
  time = 13.5,
  tz = "America/New_York"
)

MakeClockTime(
  date = "01/02/20",
  time = "13:30",
  tz = "America/New_York"
)
```

PlotByID *Plot Observed Variables by ID*

Description

The function creates one time-series plot for each observed variable. Within each plot, trajectories are overlaid by ID. Non-finite observed values, such as NaN, Inf, and -Inf, are treated as missing for plotting.

Usage

```
PlotByID(
  data,
  id,
  time,
  observed,
  ids = NULL,
  times = NULL,
  type = "b",
  pch = NULL,
  lty = NULL,
  col = NULL,
  xlab = NULL,
  ylab = NULL,
  main = NULL,
  xlim = NULL,
  ylim = NULL,
  legend = FALSE,
  ask = NULL,
  ...
)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
ids	Optional vector. ID values to plot. If NULL, all available IDs are plotted.

`times`	Optional vector of time values. If `NULL`, all available time points are plotted. If supplied, the range of `times` is used to subset the data. For example, `times = c(0, 9)` plots all rows with time values from 0 to 9, inclusive.
`type`	Character string. Plot type passed to `graphics::lines()`. Default is `"b"`.
`pch`	Optional vector of plotting characters. If `NULL`, plotting characters are generated automatically by ID. Recycled to match the number of plotted IDs.
`lty`	Optional vector of line types. If `NULL`, line types are generated automatically by ID. Recycled to match the number of plotted IDs.
`col`	Optional vector of colors. If `NULL`, colors are generated automatically by ID. Recycled to match the number of plotted IDs.
`xlab`	Optional character string. Label for the x-axis. If `NULL`, the name of the time variable is used.
`ylab`	Optional character string. Label for the y-axis. If `NULL`, the observed variable name is used.
`main`	Optional character string. Plot title prefix. If `NULL`, the observed variable name is used.
`xlim`	Optional vector of length 2. Limits for the x-axis.
`ylim`	Optional numeric vector of length 2. Limits for the y-axis. If `NULL`, limits are computed separately for each observed variable using finite observed values only.
`legend`	Logical. If `TRUE`, add a legend identifying IDs.
`ask`	Logical. If `TRUE`, ask before drawing the next observed-variable plot. The default is `TRUE` in interactive sessions when plotting more than one observed variable.
`...`	Additional arguments passed to `graphics::lines()`.

Value

Invisibly returns the IDs that were plotted.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = rep(1:4, each = 5),
  time = rep(1:5, times = 4),
  y1 = c(
    1, 2, 3, 4, 5,
    2, 3, 4, 5, 6,
    3, 4, 5, 6, 7,
    4, 5, 6, 7, 8
  ),
  y2 = c(
    5, 4, 3, 2, 1,
    6, 5, 4, 3, 2,
    7, 6, 5, 4, 3,
    8, 7, 6, 5, 4
  ),
  y3 = c(
    1, 1, 2, 2, 3,
    2, 2, 3, 3, 4,
    3, 3, 4, 4, 5,
    4, 4, 5, 5, 6
  )
)
data

PlotByID(
  data = data,
  id = "id",
  time = "time",
  observed = paste0("y", 1:3),
  ask = FALSE
)

PlotByID(
  data = data,
  id = "id",
  time = "time",
  observed = paste0("y", 1:3),
  ids = 1:3,
  times = c(1, 3),
  legend = TRUE,
  ask = FALSE
)
```

`RegularizeTimeByID` *Regularize Time by ID*

Description

The function inserts rows with missing values so that time follows a regular grid. A global grid uses the same time sequence for every ID. An ID-specific grid uses each ID's own minimum and maximum observed time values. If `method = "preserve"`, the function completes the regular grid while retaining empirical off-grid time values. If `method = "snap"`, empirical time values are snapped to the nearest grid point and the returned data use only grid time values.

Usage

```
RegularizeTimeByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  delta_t,
  grid = c("global", "by_id"),
  method = c("preserve", "snap")
)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.
`delta_t`	Positive number. Time interval.
`grid`	Character string. If `grid = "global"`, use one time grid from the global minimum to the global maximum time value. If `grid = "by_id"`, use an ID-specific time grid from each ID's minimum to maximum time value.
`method`	Character string. If `method = "preserve"`, complete the regular grid but preserve observed off-grid time values. If `method = "snap"`, snap observed time values to the nearest grid point so the output contains only regular grid times. When multiple rows for the same ID snap to the same grid point, the row closest to the grid point is kept; ties are broken by the largest number of non-missing observed/covariate values and then by original order.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), CombineByIDTime(), DeleteInitialNA(), DeleteObservedAllNA(), DeltaTByID(), DetrendByID(), DiagnoseScaleByID(), DiagnosticsByID(), DropByID(), ElapsedTimeByID(), ElapsedTimeByIDCT(), FilterByID(), FilterObservedRows(), FlagDiagnosticsByID(), GetDropID(), GetObservedInitial(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), PlotByID(), ReplaceMissingCode(), ResolveCloseTimeByID(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), ScreenByID(), SubsetByID(), SummaryByID(), TrimInitialRowsByID()

Examples

```
data <- data.frame(
  id = c(1, 1, 2, 2),
  time = c(1, 3, 1, 2),
  y = c(10, 12, 20, 21)
)
data

RegularizeTimeByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  delta_t = 1,
  grid = "by_id"
)

RegularizeTimeByID(
  data = data.frame(
    id = c(1, 1),
    time = c(1.0, 1.7),
    y = c(10, 17)
  ),
  id = "id",
  time = "time",
  observed = "y",
  delta_t = 0.5,
  grid = "by_id",
  method = "snap"
)
```

ReplaceMissingCode *Replace Missing-Value Codes*

Description

The function replaces user-specified missing-value codes with `NA` in a data frame. The replacement is applied column by column and preserves the original column classes whenever possible.

Usage

```
ReplaceMissingCode(data, values = c(-999, "-999"), columns = names(data))
```

Arguments

`data`	Data frame.
`values`	Vector. Values to replace with `NA`.
`columns`	Character vector. Names of the columns where replacement should be applied. If `NULL`, no columns are modified.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = 1:3,
  y = c(1, -999, 3),
  x = c("a", "-999", "c"),
  stringsAsFactors = FALSE
)
data

ReplaceMissingCode(
  data = data,
  values = c(-999, "-999")
)
```

`ResolveCloseTimeByID` *Resolve Close Time Points by ID*

Description

The function resolves observations that occur very close together within ID. Within each ID, rows are ordered by time and grouped into close-time clusters. Consecutive observations belong to the same cluster when the time gap between them is less than `min_gap`. One row is retained from each close-time cluster.

Usage

```
ResolveCloseTimeByID(
  data,
  id,
  time,
  observed,
  min_gap,
  method = c("max_complete", "first", "last")
)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`min_gap`	Numeric scalar or `difftime`. Minimum allowed gap between consecutive observations within ID. For numeric `time`, `min_gap` must be a positive finite numeric scalar in the same units as `time`. For `Date` or `POSIXt` `time`, `min_gap` must be a positive `difftime` object.
`method`	Character string. Method used to select the row retained from each close-time cluster. If `method = "max_complete"`, the row with the largest number of non-missing observed variables is retained. Ties are resolved by retaining the earliest row in the sorted data. If `method = "first"`, the first row in the cluster is retained. If `method = "last"`, the last row in the cluster is retained.

Details

This is useful before fitting continuous-time models, where observations that are nearly simultaneous can create numerical problems, especially when the observed values change substantially over a very small time interval.

Value

Returns a data frame with close-time observations resolved within ID.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), CombineByIDTime(), DeleteInitialNA(), DeleteObservedAllNA(), DeltaTByID(), DetrendByID(), DiagnoseScaleByID(), DiagnosticsByID(), DropByID(), ElapsedTimeByID(), ElapsedTimeByIDCT(), FilterByID(), FilterObservedRows(), FlagDiagnosticsByID(), GetDropID(), GetObservedInitial(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), PlotByID(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), ScreenByID(), SubsetByID(), SummaryByID(), TrimInitialRowsByID()

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 1, 2, 2),
  datetime = as.POSIXct(
    c(
      "2020-01-01 00:00:00",
      "2020-01-01 00:02:00",
      "2020-01-01 01:00:00",
      "2020-01-01 01:03:00",
      "2020-01-01 00:00:00",
      "2020-01-01 00:10:00"
    ),
    tz = "UTC"
  ),
  y1 = c(1, NA, 3, 4, 5, 6),
  y2 = c(NA, 2, 3, 4, 5, 6)
)

ResolveCloseTimeByID(
  data = data,
  id = "id",
  time = "datetime",
  observed = c("y1", "y2"),
  min_gap = as.difftime(
    5,
    units = "mins"
  ),
  method = "max_complete"
)
```

ResolveDuplicateIDTime *Resolve Duplicate ID-Time Rows*

Description

The function resolves exact duplicate `id-time` rows using a simple deterministic rule. This is useful before inserting missing rows on a time grid because grid-completion functions generally require unique `id-time` combinations.

Usage

```
ResolveDuplicateIDTime(
  data,
  id,
  time,
  observed = NULL,
  covariates = NULL,
  method = c("first", "last", "max_complete"),
  order_by = NULL,
  decreasing = FALSE
)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.
`method`	Character string. Rule used to choose one row from each duplicated `id-time` group. If `method = "first"`, keep the first row after sorting by `id` and `time`. If `method = "last"`, keep the last row after sorting by `id` and `time`. If `method = "max_complete"`, keep the row with the largest number of non-missing values in `observed`.
`order_by`	Character vector. Optional columns used to break ties after the main `method` rule.
`decreasing`	Logical. If `TRUE`, sort `order_by` columns in decreasing order.

Value

Returns a data frame with unique `id-time` rows.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = c(1, 1, 1),
  time = c(0, 0, 1),
  y1 = c(NA, 1, 2),
  y2 = c(NA, 1, 3)
)
data

ResolveDuplicateIDTime(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  method = "max_complete"
)
```

`RoundClockTime` *Round Clock Time*

Description

The function rounds, floors, or ceilings clock times to a specified unit. It supports `POSIXt`, `Date`, and numeric time values.

Usage

```
RoundClockTime(
  x,
  unit = "hour",
  method = c("round", "floor", "ceiling"),
  origin = "1970-01-01",
  tz = "UTC"
)
```

Arguments

`x`	A `POSIXt`, `Date`, or numeric vector.
`unit`	Character string. Unit to round to. Supported values are `"second"`, `"minute"`, `"hour"`, `"day"`, and their plural forms.
`method`	Character string. One of `"round"`, `"floor"`, or `"ceiling"`.
`origin`	Origin for numeric time values. Only used if `x` is numeric.
`tz`	Character string. Time zone used when numeric time values are converted to `POSIXct`.

Value

Returns a vector with rounded times.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
x <- as.POSIXct(
  "2020-01-01 10:31:00",
  tz = "America/New_York"
)

RoundClockTime(
  x = x,
  unit = "hour"
)
```

ScaleByID *Scale by ID*

Description

The function scales the data by ID.

Usage

```
ScaleByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  scale = TRUE,
  obs_skip = NULL,
  cov_skip = NULL
)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.
`scale`	Logical. If `scale = TRUE`, standardize by `id`. If `scale = FALSE`, mean center by `id`.
`obs_skip`	Character vector. A vector of character strings of the names of the observed variables to skip centering/scaling.
`cov_skip`	Character vector. A vector of character strings of the names of the covariates to skip centering/scaling.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), CombineByIDTime(), DeleteInitialNA(), DeleteObservedAllNA(), DeltaTByID(), DetrendByID(), DiagnoseScaleByID(), DiagnosticsByID(), DropByID(), ElapsedTimeByID(), ElapsedTimeByIDCT(), FilterByID(), FilterObservedRows(), FlagDiagnosticsByID(), GetDropID(), GetObservedInitial(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), PlotByID(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveCloseTimeByID(), ResolveDuplicateIDTime(), RoundClockTime(), ScreenByID(), SubsetByID(), SummaryByID(), TrimInitialRowsByID()

Examples

```
data <- data.frame(
  id = rep(1:3, each = 4),
  time = rep(1:4, times = 3),
  y1 = c(
    1, 2, 3, 4,
    10, 11, 12, 13,
    20, 22, 24, 26
  ),
  y2 = c(
    4, 3, 2, 1,
    13, 12, 11, 10,
    26, 24, 22, 20
  )
)
data

ScaleByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)

ScaleByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  scale = FALSE
)
```

ScreenByID *Screen IDs Using Dynamic-Data Diagnostics*

Description

The function computes ID-level diagnostics, applies rule-based flags, and returns IDs that are candidates for exclusion or sensitivity analysis. It is a wrapper around `DiagnosticsByID()`, `FlagDiagnosticsByID()`, and `GetDropID()`.

Usage

```
ScreenByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  missing_codes = NULL,
  min_nonmissing = 1L,
  extreme_cut = c(4, 5, 6),
  sd_cut = c(0.1, 0.05),
  time_scale = 1,
  posix_unit = "hours",
  min_observed_rows = 30L,
  min_complete_rows = NULL,
  max_prop_all_missing = 0.95,
  max_gap = NULL,
  max_median_gap = NULL,
  min_sd = 0.05,
  flag_extreme_cut = 6,
  drop_score_cut = 2L,
  flagged_only = TRUE
)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.

`covariates` Character vector. A vector of character strings of the names of the covariates in the data.

`missing_codes` `NULL` or vector. Values to temporarily treat as missing when computing diagnostics. The input data are not modified.

`min_nonmissing` Positive integer. Minimum number of finite observed variables required for a row to count as an observed row.

`extreme_cut` Numeric vector. Absolute-value cutoffs used to count extreme values.

`sd_cut` Numeric vector. Within-ID standard-deviation cutoffs used to count low-variance variables.

`time_scale` Positive number. Multiplier applied to numeric time gaps. For example, use `time_scale = 24` when time is measured in days and gaps should be reported in hours.

`posix_unit` Character string. Unit for gaps when `time` is a `POSIXt` or `Date` variable. Passed to `as.numeric.difftime()`.

`min_observed_rows`
Positive integer. Minimum number of observed rows expected per ID.

`min_complete_rows`
`NULL` or positive integer. Minimum number of complete rows expected per ID.

`max_prop_all_missing`
`NULL` or number between 0 and 1. Maximum tolerated proportion of all-missing observed rows.

`max_gap` `NULL` or positive number. Maximum tolerated observed-row time gap in the units used by `DiagnosticsByID()`.

`max_median_gap` `NULL` or positive number. Maximum tolerated median observed-row time gap in the units used by `DiagnosticsByID()`.

`min_sd` `NULL` or non-negative number. Minimum tolerated within-ID standard deviation across observed variables. If `NULL`, low-SD flagging is skipped.

`flag_extreme_cut`
Positive number. Absolute-value cutoff used by `FlagDiagnosticsByID()`. This value is added to `extreme_cut` before calling `DiagnosticsByID()` so that the required total-count column is available.

`drop_score_cut` Positive integer. IDs with priority scores greater than or equal to this value are marked as sensitivity-drop candidates.

`flagged_only` Logical. If `TRUE`, return IDs marked as sensitivity-drop candidates. If `FALSE`, return all flagged IDs.

Value

Returns a list with the following elements:

`diagnostics` Diagnostics with flag columns appended.

`drop_id` IDs selected by `GetDropID()`.

`keep_id` IDs not in `drop_id`.

`call` The matched call.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), CombineByIDTime(), DeleteInitialNA(), DeleteObservedAllNA(), DeltaTByID(), DetrendByID(), DiagnoseScaleByID(), DiagnosticsByID(), DropByID(), ElapsedTimeByID(), ElapsedTimeByIDCT(), FilterByID(), FilterObservedRows(), FlagDiagnosticsByID(), GetDropID(), GetObservedInitial(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), PlotByID(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveCloseTimeByID(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), SubsetByID(), SummaryByID(), TrimInitialRowsByID()

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 3, 1, 2, 3),
  y1 = c(1, NA, 3, 1, 1, 1),
  y2 = c(NA, NA, 4, 2, 2, 2)
)

ScreenByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  min_observed_rows = 3,
  min_sd = 0.05
)
```

SubsetByID *Subset Data Set by ID*

Description

The function creates a list of data frames for each ID.

Usage

```
SubsetByID(data, id, time, observed, covariates = NULL)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.

`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.

Value

Returns a list by ID numbers.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveCloseTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = rep(1:3, each = 4),
  time = rep(1:4, times = 3),
  y1 = c(
    1, 2, 3, 4,
    10, 11, 12, 13,
    20, 21, 22, 23
  ),
  y2 = c(
    4, 3, 2, 1,
    13, 12, 11, 10,
    23, 22, 21, 20
  )
)
data

SubsetByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)
```

SummaryByID *Summarize Dynamic Modeling Data by ID*

Description

The function returns a simple diagnostic table with one row per ID. For model-screening workflows, prefer `DiagnosticsByID()`, `FlagDiagnosticsByID()`, and `ScreenByID()`.

Usage

```
SummaryByID(data, id, time, observed, covariates = NULL)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.

Details

The `initial_na` column is based on the observed variables only. Covariates are retained in the input checks and sorted data, but missing covariates in the first row do not cause `initial_na = TRUE`.

Value

Returns a data frame with one row per ID.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnoseScaleByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `ElapsedTimeByIDCT()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, LagByID(), MakeClockTime(), PlotByID(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveCloseTimeByID(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), ScreenByID(), SubsetByID(), TrimInitialRowsByID()

Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = rep(1:3, times = 2),
  y1 = c(1, NA, 3, 4, 5, 6),
  y2 = c(1, 2, 3, NA, NA, NA)
)
data

SummaryByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)
```

`TrimInitialRowsByID` *Trim Leading Rows With Too Few Observed Values by ID*

Description

The function removes leading rows within each ID until the first retained row has at least `min_nonmissing` non-missing observed variables. This is useful before creating an elapsed-time variable when leading records contain no usable measurements. Unlike `DeleteInitialNA()`, the default does not require the first retained row to be complete on all selected variables.

Usage

```
TrimInitialRowsByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  min_nonmissing = 1L
)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.

id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.
min_nonmissing	Positive integer. Minimum number of non-missing observed variables required in the first retained row for each ID.

Value

Returns a data frame.

Author(s)

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See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), CombineByIDTime(), DeleteInitialNA(), DeleteObservedAllNA(), DeltaTByID(), DetrendByID(), DiagnoseScaleByID(), DiagnosticsByID(), DropByID(), ElapsedTimeByID(), ElapsedTimeByIDCT(), FilterByID(), FilterObservedRows(), FlagDiagnosticsByID(), GetDropID(), GetObservedInitial(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), PlotByID(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveCloseTimeByID(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), ScreenByID(), SubsetByID(), SummaryByID()

Examples

```
data <- data.frame(
  id = rep(1:2, each = 4),
  time = rep(1:4, times = 2),
  y1 = c(NA, 1, 2, 3, NA, NA, 1, 2),
  y2 = c(NA, NA, 2, 3, NA, 1, 1, 2)
)

TrimInitialRowsByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  min_nonmissing = 1
)
```

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